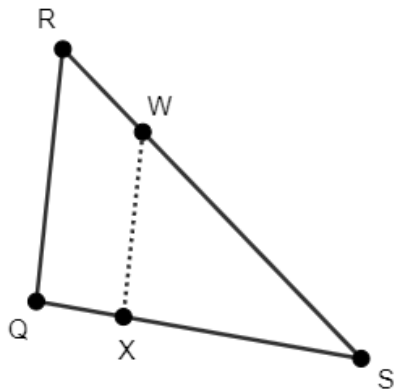


$\triangle QRS$ is shown, where $m\angle QRS = 50^\circ$, $m\angle RQS = 94^\circ$, and $m\angle RSQ = 36^\circ$. Point W is on $\overline{RS}$ and Point X is on $\overline{QS}$. $\overline{WX}$ is drawn as a dotted line.



1. Use a ruler and a protractor to complete the table.

Measurements of $\triangle QRS$		Measurements of $\triangle XWS$	
$QR = 3.3 \text{ cm}$	$m\angle QRS = 50^\circ$	$XW = 2.4 \text{ cm}$	$m\angle XWS = 50^\circ$
$RS = 5.8 \text{ cm}$	$m\angle RQS = 94^\circ$	$WS = 4.2 \text{ cm}$	$m\angle WXS = 94^\circ$
$QS = 4.4 \text{ cm}$	$m\angle RSQ = 36^\circ$	$XS = 3.2 \text{ cm}$	$m\angle WSX = 36^\circ$

2. The corresponding angles of $\triangle QRS$ and $\triangle XWS$ are **congruent**.

3. Complete the table.

Sides	Ratio
Determine the ratio of $\overline{QR}$ to $\overline{XW}$.	11:8
Determine the ratio of $\overline{RS}$ to $\overline{WS}$.	11:8
Determine the ratio of $\overline{QS}$ to $\overline{XS}$.	11:8

4. The corresponding sides of $\triangle QRS$ and $\triangle XWS$ are **proportional**.

5. $\triangle QRS$ is similar to $\triangle XWS$.
6. Describe a sequence of transformations that can be used to map $\triangle QRS$ onto $\triangle XWS$.
Dilate $\triangle QRS$ by a scale factor of $\frac{8}{11}$ about Point S .

Complete each relationship for $\triangle QRS$ and $\triangle XWS$.

7. $\frac{RS}{WS} = \frac{QS}{XS}$

8. $\frac{WX}{RQ} = \frac{XS}{QS}$

9. $\frac{RQ}{WX} = \frac{RS}{WS}$

10. Observing that $\frac{RQ}{WX} \neq \frac{RW}{WS}$, your classmate Charlee claims that $\triangle QRS$ and $\triangle XWS$ are not similar. If her logic is faulty, explain why.

$\triangle QRS$ and $\triangle XWS$ are similar because side $\overline{RS}$ corresponds to side $\overline{WS}$. Charlee should have used the equation $\frac{RQ}{WX} = \frac{RS}{WS}$ to determine triangle similarity.